Shirpur Education Society's

R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR

An Autonomous Institute

(Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere)

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(स्वायत्त महाविद्यालय)

Programme: B.Tech in Artificial Intelligence (AI) and Data Science

Year: II/Semester III (Exam Year: 2025-2026)

Subject: Database Management Systems **Time:** 02:30 pm - 04:30 pm (02:00 Hrs.)

Date: 04 Dec 2025 **Max Marks:** 60

END SEMESTER EXAMINATION ODD SEM III-2025-REGULAR

Instructions:
1. This question paper contains 3 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks

1. **15**

A. 1. **10**

i. Analyze the requirements of a Hostel Management System and identify the major entities, attributes, and relationships. Based on your analysis, justify and draw an appropriate ER Diagram for the system (include 6-7 entities). **10**

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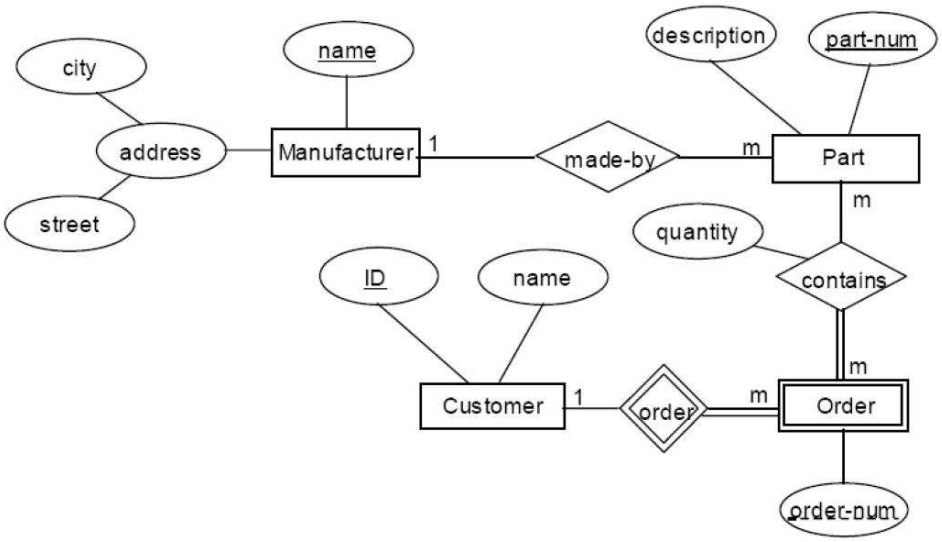
ii. Write the steps to convert ER to Relational Tables and Convert the following ER into Relational Model **10**



B. List the properties of B-Tree and create a B-tree of order 3 by inserting the following elements: 78,52,81,40,33,90,85,20,38 **5**

2. **15**

A. Write Data control commands (DCL) and transaction control commands (TCL) in SQL with one example for each. **5**

B. 2. **10**

i. What is view serializability? Explain the conditions of view equivalence schedule. Check whether the given schedule S is view serializable or not. If yes, then give the serial schedule. S: R1(A), W2(A), R3(A), W1(A), W3(A). **10**

OR

ii. Explain Validation-Based (Optimistic) Concurrency Control. Describe the three phases of validation and show with an example how conflicts are detected during validation **10**

3. **15**

A. 1. **5**

i. Relation R(X,Y,Z,W,L) with FDs: **5**

F={X→Y, Y→Z, {X,L}→W}.

Calculate Key and normalize it using 1NF,2NF, 3NF and BCNF

---------- OR ----------

ii. Explain partial and transitive dependencies with examples. **5**

B. Explain the role and responsibilities of a Database Administrator (DBA). **5**

C. Explain ACID properties of a transaction with suitable examples. **5**

4. **15**

A. 1. **10**

i. Write SQL queries for following **10**

CUSTOMER (CustomerID, CustomerName, City, PhoneNo)

PRODUCT (ProductID, ProductName, Category, Price, SupplierID)

ORDERS (OrderID, CustomerID, ProductID, OrderDate, Quantity, TotalAmount)

SUPPLIER (SupplierID, SupplierName, City)

i) Modify the price of the product with ProductID = 105 to ₹2,500.

ii) List all products belonging to the 'Electronics' category in descending order.

iii) Display the customer name, product name, and order date for all orders.

iv) Find the products whose price is higher than the average price of all products.

v) Create a VIEW named HighValueOrders to display CustomerName, ProductName, and TotalAmount for orders where the total amount exceeds ₹10,000.

----------OR ----------

ii. 1. Student(Stud_ID, Stud_Name, Dept) **10**

i. Create a procedure to insert an entry of new student in student table.

ii. Grant SELECT and UPDATE privileges on Student table to user exam, admin.

iii. Revoke UPDATE permission on STUDENT from user exam.

iv. Compare functions and procedures

B. A hash table has size 10 and uses linear probing to resolve collisions. Given the following input 721, 234, 147, 671, 174, 627, 173, 241; create the hash table. **5**